ESO208 COMPUTATIONAL METHODS IN ENGINEERING

ASSIGNMENT 9

MAXIMUM MARKS: 30

DUE DATE: 18 OCTOBER 2026

Please upload hand-written solutions on Gradescope. Typed solutions except codes and plots will not be graded. Please upload single PDF file for hand-written part.

Q.1. [15 marks] The motion of a damped spring-mass system is described by the following ODE:

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$$

where $x = \text{displacement from equilibrium position } (m), t = \text{time } (s), m = 20\ kg, c = \text{damping coefficient, } \mathrm{k} = \text{spring constant} = 20\mathrm{N/m}.$

(a) Convert the second order ODE into a system of first order ODEs and write the equations for the 4th order RK method (hand-written). [5]

(b) Write a code to solve the system of first order ODEs for $0 \leq t \leq 15\ s$ at 1 s interval if $x(t=0) = 1$ and $x'(t=0) = 0$ for $c = 5, 40\ and\ 60\ Ns/m$ using the fourth order RK method. [7]

(b) Plot the displacement as a function time for the three damping coefficients on the same graph. [3]

Q.2. [10 marks] The following differential equation needs to be solved for $0 \leq t \leq 1$ using a step size of 0.1.

$$\frac{dy}{dt} = -10y$$

$$y(0) = 1$$

(a) Write a code to solve the ODE using explicit and implicit Euler method. [7]

(b) Make a plot of y versus t for the analytical solution and numerical solutions from explicit and implicit Euler schemes on the same graph. [3]

Q.3. [5 marks] The following ODE needs to be solved:

$$\frac{dy}{dt} = -\frac{2y}{t}$$

The values of y at four points are known: $y(2.5) = 0.4800, y(3) = 0.3333, y(3.5) = 0.2449$ and $y(4) = 0.1875$. Predict the value of $y(4.5)$ using the fourth order Adam's method using a tolerance $\varepsilon_s = 0.01\%$ (hand-written).